

Amazon Sales Analysis 2023

Introduction

Amazon Sales Analysis 2023 is a project developed to analyse Sales from Amazon website in the year 2023. We have extracted data from Kaggle and after processing it, we have built visuals to generate insights.

Dataset

[Amazon Sales Dataset](#) is taken from [Kaggle](#), which was initially scrapped from the Amazon website. The raw dataset consists of 140 csv files, categorized by product categories on Amazon website. Following image shows a snap of raw dataset.

Air Conditioners.csv (247.75 kB)

Detail

Compact

Column

9 of 9 columns

About this file

This CSV file is having the data 500+ Air Conditioners Product's Details

name	main_category	sub_category	image	link	ratings
The name of the product					
708 unique values	1 unique value	1 unique value	635 unique values	720 unique values	[null] 4.0 Other (37)
Lloyd 1.5 Ton 3 Star Inverter Split Ac (5 In 1 Convertible, Copper, Anti-Viral + Pm 2.5 Filter, 2023...	appliances	Air Conditioners	https://m.media-amazon.com/images/I/31UISB98sYL._AC_UL320_.jpg	https://www.amazon.in/Lloyd-Inverter-Convertible-Anti-Viral-GLS1813FWAMC/dp/B0BRKXTS8T/ref=sr_1_4?q1...	4.2
LG 1.5 Ton 5 Star AI DUAL Inverter Split AC (Copper, Super Convertible 6-in-1 Cooling, HD Filter wit...	appliances	Air Conditioners	https://m.media-amazon.com/images/I/51JFb7FctDL._AC_UL320_.jpg	https://www.amazon.in/LG-Convertible-Anti-Virus-Protection-RS-Q19YNZE/dp/B0BQ3MXML8/ref=sr_1_5?qid=1...	4.2
LG 1 Ton 4 Star Ai Dual Inverter Split	appliances	Air Conditioners	https://m.media-amazon.com/images/I/	https://www.amazon.in/LG-Inverter-	4.2

Data Explorer

374.41 MB

Air Conditioners.csv

All Appliances.csv

All Books.csv

All Car and Motorbike Produc

All Electronics.csv

All English.csv

All Exercise and Fitness.csv

All Grocery and Gourmet Foo

All Hindi.csv

All Home and Kitchen.csv

All Movies and TV Shows.csv

All Music.csv

All Pet Supplies.csv

All Sports Fitness and Outdoc

All Video Games.csv

Amazon Fashion.csv

Amazon Pharmacy.csv

Amazon-Products.csv

Baby Bath Skin and Grooming

Baby Fashion.csv

Baby Products.csv

Backpacks.csv

Badminton.csv

Bags and Luggage.csv

Each csv file have ten columns which are as follows:

- ❖ name: Name of the product
- ❖ main_category: Main category of the product
- ❖ sub_category: Sub category of the product
- ❖ image: Image of the product look like
- ❖ link: Amazon website link of the product
- ❖ ratings: Ratings given by Amazon customers to the product
- ❖ no_of_rating: Number of ratings given to the product
- ❖ discount_price: Price of the product after discount
- ❖ actual_price: Actual price of the product

Data Processing

Python (pandas) is used to process the data, and Python (os) is used to automate the process. The script starts with the first file in the directory and runs till the last file, and performs following operations for each file:

- Reads the csv file, and converts it into a dataframe
- Drops unwanted columns, i.e., name, image, and link
- Replace all NULL values with zero in rating, no_of_rating, discount_price, and actual_price columns.
- Clean rating, no_of_rating, discount_price, and actual_price columns for non-digit (unwanted) characters.
- Merge cleaned dataframe in a single dataframe.

The script then converts the final dataframe to a csv file, which is the final dataset which is used further in the project.

Data Insights

Using Tableau, two dashboards are built to generate insights.

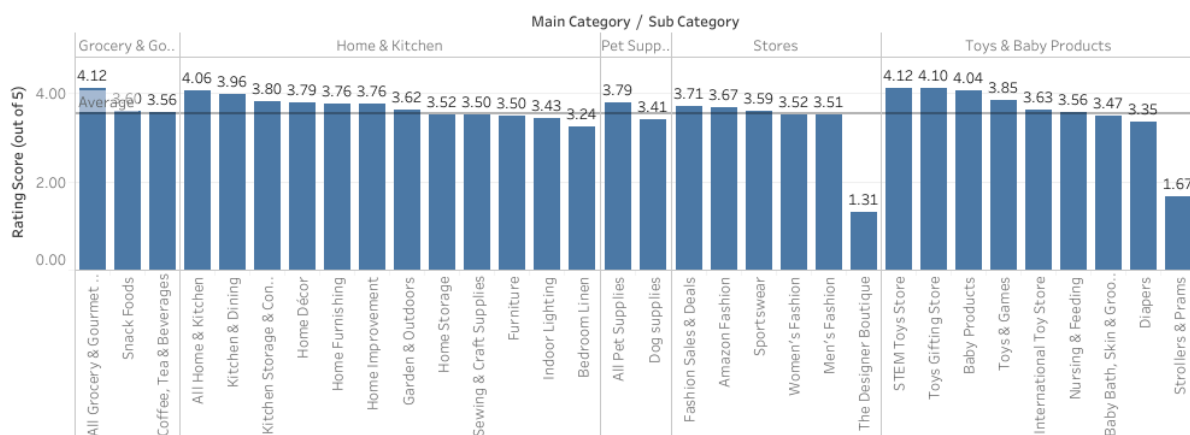
Amazon Sales 2023 (1)

Total Products Sold: 1,048,576 INR

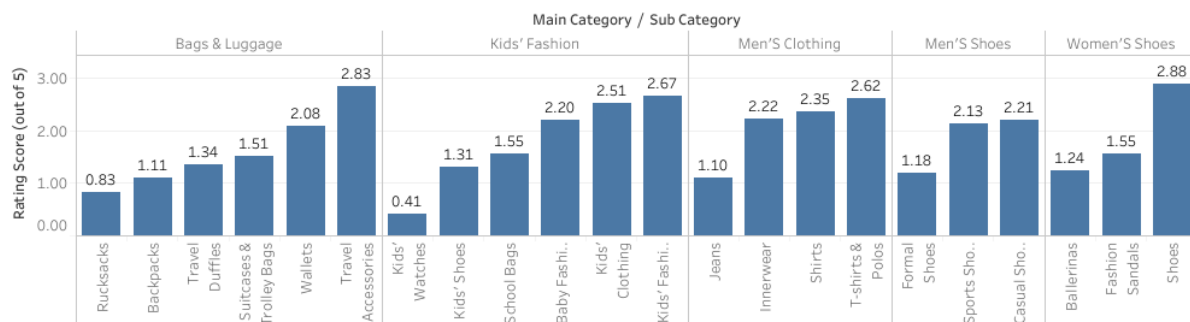
Total Sales: 2,060,979,361 INR

Total Savings: 2,462,890,578 INR

Most Rated Categories



Least Rated Categories



Below is the elaboration of the above dashboard.

Top 20 Most Rated Categories

Table below shows the top 20 most rated sub-categories along with their ratings and the main-category in which they lie

Toys & Baby Products	4.12 4.10 4.04 3.85 3.63 3.56	STEM Toy Store Toys Gifting Store Baby Products Toys & Games International Toy Store Nursing & Feeding
Grocery & Gourmet Foods	4.12 3.60 3.56	All Grocery & Gourmet Foods Snack Foods Coffee, Tea & Beverages
Home & Kitchen	4.06 3.96 3.80 3.79 3.76 3.76 3.62	All Home & Kitchen Kitchen & Dining Kitchen Storage & Containers Home & Decor Home Furnishing Home Improvement Garden & Outdoors
Pet Supplies	3.79	All Pet Supplies
Stores	3.71 3.67 3.59	Fashion Sales & Deals Amazon Fashion Sportswear

Top 20 Least Rated Categories

Table below shows the top 20 least rated sub-categories along with their ratings and the main-category in which they lie.

Kid's Fashion	0.41	Kid's Watches
Bags & Luggage	0.83 1.11 1.34 1.51 2.08	Rucksacks Backpacks Travel Duffles Suitcases & Trolley Bags Wallets
Men's Clothing	1.10 2.22 2.35	Jeans Innerwear Shirts
Men's Shoes	1.18 2.13 2.21 2.62	Formal Shoes Sports Shoes Casual Shoes T-Shirts & Polos
Women's Shoes	1.24	Ballerinas

	1.55	Fashion Sandals
Kid's Fashion	1.31	Kid's Shoes
	1.55	School Bags
	2.20	Baby Fashion
	2.51	Kid's Clothing
	2.67	Kid's Fashion

Dashboard below - Amazon Sales 2023 (2) shows three more insights.

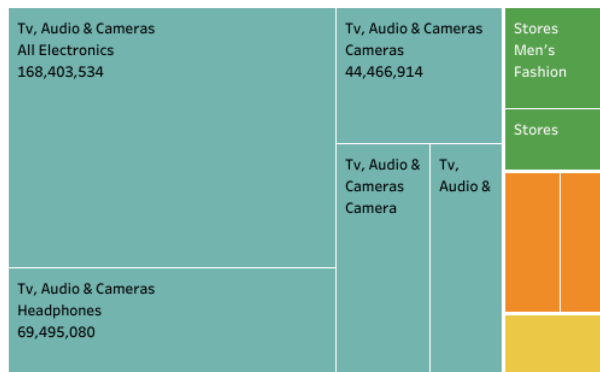
Amazon Sales 2023 (2)

Total Products Sold: 1,048,576 INR

Total Sales: 2,060,979,361 INR

Total Savings: 2,462,890,578 INR

Most Popular Categories



Price vs Category



Least Popular Categories



Main Category
■ Appliances
■ Stores
■ Toys & Baby Products
■ Tv, Audio & Cameras

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Appendix - A

Following is the script that is used for data processing

```
import os
import pandas as pd
import numpy as np
import csv

def getDataFileNames(folderPath):
    fileNames = []

    try:
        if not os.path.isdir(folderPath):
            print("Error: Folder path doesn't exist. Please check and try again.")
            return

        for file in os.listdir(folderPath):
            if os.path.isfile(os.path.join(folderPath, file)):
                fileNames.append(file)

        return fileNames

    except Exception as e:
        print("Error:" + e)
        return

folderPath = r"D:\DATA Projetics\Amazon Sales\amazon\Dataset"
dataFileNames = getDataFileNames(folderPath)
print(dataFileNames)

def dataCleaning(df, dataFileName):
    try:
        dfCopy = df.copy()
        colsToModify = ['ratings', 'no_of_ratings', 'discount_price', 'actual_price']

        #Dropping unwanted columns
        dfCopy = dfCopy.drop(columns=["name", "image", "link"], axis=1)

        dfCopy['main_category'] =
dfCopy['main_category'].astype(str).str.title()
```

```

        for col in colsToModify:
            #Replacing all NULL values with '0'
            dfCopy[col] = dfCopy[col].fillna(0)

            #Removing non-digit characters
            dfCopy[col] =
dfCopy[col].astype(str).str.replace(r'^0-9.', '', regex=True)

        return dfCopy

    except Exception as e:
        print(f"Unexpected error while cleaning data -> {e}")
        return

try:
    finalDF = pd.DataFrame()

    for dataFileName in dataFileNames:
        df = pd.read_csv(os.path.join(folderPath, dataFileName))
        cleanedDF = dataCleaning(df, dataFileName)

        if cleanedDF is not None:
            print(f"-"*10)
            print(f"Data cleaned for '{dataFileName}' successfully")
            finalDF = pd.concat([finalDF, cleanedDF],
ignore_index=True)
            print(f"Data added for '{dataFileName}'")
        else:
            print(f"ERROR -> Function returned None for
'{dataFileName}'")
            break

    finalDF.to_csv("cleanedDataset7.csv", index=False)
    print(f"Cleaned Dataset created successfully")

except Exception as e:
    print(f"Unexpected error -> {e}")

```